



Alberta Public Health Disease Management Guidelines

Ebola Virus Disease



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Alberta Health

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Revisions

Revision Date	Document Section	Description of Revision
December 2023	General	Entire guideline updated to align with current information
	Case Definition	Added clarity to distinguish between probable and suspect cases
	Reporting Requirements	Included Canada Border Services Agency and International Health Regulation
	Post-exposure prophylaxis	Included information on Ebola Zaire vaccine authorization for use in Canada

Case Definition

Confirmed Case

Laboratory confirmation of Ebola virus infection performed using at least one of the methods below:

- Isolation and identification of virus from an appropriate clinical specimen (e.g., blood, serum, tissue, urine specimens or throat secretions) (performed at the National Microbiology Laboratory [NML]);
OR
- Detection of virus-specific RNA by reverse-transcriptase (RT-PCR) from an appropriate clinical specimen (e.g., blood, serum, tissue) using two independent targets or two independent samples AND confirmed by NML by nucleic acid testing or serology,
OR
- Demonstration of virus antigen in tissue (e.g., skin, liver or spleen) by immunohistochemical or immunofluorescent techniques AND another test e.g., PCR,
OR
- Demonstration of specific IgM AND IgG antibody by EIA, immunofluorescent assay or Western Blot by the NML or an approved World Health Organization (WHO) collaboration centre,
OR
- Demonstration of seroconversion or a fourfold rise in IgG serum antibody by EIA, immunofluorescent assay or Western Blot from an acute vs a convalescent serum sample.

Probable Case

- Any suspect case evaluated by a clinician or any deceased suspect case, where it is not possible to collect specimens for laboratory confirmation.

Suspect Case

- A person with clinical illness^(A) **AND** at least one of the following epidemiologic risk factors within the 21 days before the onset of symptoms in whom EVD has not been ruled out:
 - sexual contact with a probable or confirmed EVD case,
 - health care worker (HCW)^(B) and/or household carer^(C), who directly or indirectly cared for a probable or confirmed case of EVD (e.g., direct patient care or contact with environment or fomites of a case),
 - household member of a probable or confirmed case of EVD,
 - close contact^(D) in a community setting with a probable or confirmed case of EVD while the person was symptomatic,
 - laboratory worker who processed body fluids or tissues of a probable or confirmed EVD case,
 - direct exposure to human remains (e.g., through participation in funeral or burial rites) in a geographic area where an EVD outbreak is occurring,
 - direct exposure to human remains of a probable or confirmed EVD case,
 - direct contact with bats, primates or wild bush meat from a currently active EVD-affected area

OR

- A person with clinical illness^(A)
 - AND**
 - residence in or travel to an area where EVD transmission is active within the 21 days before symptom onset
 - AND**
 - has a history of close contact with an ill individual while in that area
 - AND**
 - is determined to be highly suspicious for Ebola by Public Health.

^(A) Clinical illness is defined as an individual presenting with at least one of the following symptoms:

- fever (temperature ≥ 38.0 degrees Celsius) or subjective fever
- malaise
- myalgia
- headache
- arthralgia
- fatigue
- loss of appetite
- conjunctival redness
- sore throat
- chest pain
- abdominal pain
- nausea
- vomiting
- diarrhea that can be bloody
- hemorrhage
- erythematous maculopapular rash on the trunk

^(B) Health care Worker (HCW) is defined as:

- all health practitioners (any person who provides health care or treatment to any person) and
- all individuals at increased risk for exposure to, and/or transmission of, a communicable disease because they work, study, or volunteer, in one or more of the following health care environments:
 - hospital
 - nursing home, supportive living accommodation, or home care setting,
 - mental health facility,
 - community setting,
 - office or clinic of a health practitioner,
 - clinical laboratory.

^(C) Household carers perform similar functions as health care workers, but in non-health care settings.

^(D) Close contact is defined as being within two meters of a confirmed or probable case for a prolonged period of time.

Reporting Requirements

Physicians, Health Practitioners and Others

- Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone of all confirmed, probable or suspect EVD cases by the Fastest Means Possible (FMP).
 - Laboratory testing should only be conducted following consultation with the MOH and Virologist on call.

Laboratories

- All laboratories shall report all laboratory results for Ebola virus tests, including negatives, by FMP to the:
 - MOH (or designate) of the zone, and
 - Chief Medical Officer of health (CMOH) (or designate)

Alberta Health Services and First Nations and Inuit Health Branch

- The MOH (or designate) where the case currently resides shall notify the CMOH (or designate) by FMP of all confirmed, probable and suspect cases.
- The MOH (or designate) where the case currently resides shall forward:
 - The preliminary Public Health Agency of Canada's [Ebola Virus Disease Case Report Form](#) of all confirmed, probable and suspect cases to the CMOH (or designate) immediately upon notification.
 - updated forms should be sent to the CMOH (or designate) when more information becomes available, including if the individual is determined NOT to be a case
 - The initial Notifiable Disease Report (NDR) of all confirmed and probable cases to the CMOH (or designate) within one week of notification and the final NDR (amendment) within two weeks of notification.
- For out-of-province and out-of-country cases/contacts, the following information should be forwarded to the CMOH (or designate) by FMP:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - attending physician (locally and out-of-province) and
 - positive laboratory report (if applicable), and
 - other relevant clinical/epidemiological information.

Canada Border Services Agency (CBSA)

- Travelers from EVD-affected areas with symptoms, or who have had close contact with a person infected with Ebola are referred to Public Health Agency of Canada (PHAC) Quarantine Officers (QO) for health screening.
 - Travelers are reported to the provincial or territorial public health authority under subsection 25(1) of the *Quarantine Act*.
- The Quarantine Officer (QO) will determine the most appropriate response in consultation with the MOH, based on the following assessment of the traveller:
 - symptomatic and at risk of having Ebola, direct to a hospital for medical attention.
 - close contact exposure but does not pose an immediate risk of significant harm to public health, immediately self-isolate and contact Health Link at 811 should EVD-compatible symptoms develop.

International Health Regulation Reporting

- Under the [*International Health Regulations \(IHR\) \(2005\)*](#), enhanced reporting of Ebola to the World Health Organization (WHO) is mandatory. Alberta Health notifies the Public Health Agency of Canada (PHAC) of every confirmed, probable, or suspect case of Ebola and PHAC notifies the WHO of all confirmed cases within 24 hours of initial notification.

Clinical Assessment and Epidemiology

Etiology

Ebola virus disease (EVD), is caused by viruses within the genus *Orthoebolavirus* (formerly *Ebolavirus*) a member of the Filoviridae family.^(1,2) There are six species of *orthoebolavirus* named based on where they were originally discovered. They are listed below along with their common names⁽¹⁾.

- *Orthoebolavirus zairense* (Ebola virus, EBOV),
- *Orthoebolavirus sudanense* (Sudan virus, SUDV),
- *Orthoebolavirus taiense* (Tai Forest virus, TAFV),
- *Orthoebolavirus bundibugyoense* (Bundibugyo virus, BDBV)
- *Orthoebolavirus restonense* (Reston virus, RESTV)
- *Orthoebolavirus bombaliense* (Bombali virus, BOMV)

Out of the six species only four species (Sudan, Bundibugyo, Tai Forest and Zaire ebolavirus) have been known to cause illness in humans. To date no human illness caused by of Reston and Bombali have been reported.⁽³⁾

Clinical Presentation

There are three stages:⁽⁴⁾

Prodromal stage: sudden onset of flu like symptoms (fever, malaise, myalgia, sore throat, and headache) followed by worsening weakness, anorexia, diarrhea, nausea and vomiting. Duration can be up to 10 days.

Second stage: characterized by symptoms and clinical manifestations that affect many organ systems which can include:

- gastrointestinal (vomiting, diarrhea, anorexia, and abdominal pain)
- neurological (headaches and confusion)
- vascular (conjunctival; pharyngeal)
- cutaneous (maculopapular rash)
- respiratory (cough, chest pain, shortness of breath)
- complete exhaustion
- hemorrhagic manifestations may also appear during this stage and may be internal and external.

End stage: People with fatal disease typically present with more severe clinical symptoms in the early stages and succumb to the illness between day six and day 16. This is a result of a combination of multi-organ failure and hypovolemic shock due to severe fluid loss.^(5,6) People that survive EVD, may have a fever for several days and begin to improve around day six.⁽⁶⁾ From past outbreaks, the case fatality rate varies from 25%–90%.⁽⁷⁾ Pregnant individuals are at an increased risk of adverse outcomes, including fetal loss and pregnancy-associated hemorrhage.⁽⁸⁾

Post EVD sequelae among survivors decrease over time but may persist for 48 months or longer after recovery. Symptoms may involve one or more of the following: abdominal (abdominal or pelvic pain, gastritis), neurological (headache, dizziness), ocular (conjunctivitis, iridocyclitis, cataract, vision problem, ocular pain), musculoskeletal (myalgia; neck, back and joint pain), and general symptoms (fever, fatigue, anorexia).^(9–11)

Diagnosis

Any laboratory testing of an individual with suspected EVD must be done in consultation with the MOH of the zone and the Virologist/Microbiologist on call. The diagnosis can be confirmed by one or more of the following diagnostic markers:

- direct detection of virus in blood, serum, tissue, urine specimens or throat secretions
- blood or tissue samples by virus isolation,
- presence of virus-specific RNA by RT-PCR
- enzyme-linked immunosorbent assay (ELISA) for viral antigens or immunoglobulin (IgM)
- immunohistochemistry on formalin-fixed tissues
- presence of virus-specific immunoglobulin class M antibodies or rising immunoglobulin class G antibody tiers.

The gold standard for confirmation of infection is a PCR positive result by two molecular targets.⁽¹⁰⁾ As *ebolavirus* may not be detectable in blood within the first three days after the onset of fever, virus isolation and RT-PCR tests may be negative if testing is done before this three-day threshold. To reliably rule out EVD, repeat testing three days after onset of fever should be considered.⁽⁹⁾

Ebola RT-PCR testing is available at the [Alberta Precision Laboratory](#) (APL), a containment level 3 facility. Positive Ebola RT-PCR results from APL require confirmation from the NML. Negative Ebola RT-PCR results from APL should be assessed in the overall clinical and epidemiologic context of a particular case for a decision as to whether the patient should be removed from isolation or not. The MOH-clinical-lab team may make this decision prior to receipt of a confirmatory Ebola negative result from APL.

- No virus culture should be attempted outside of NML, a containment level 4 laboratory.⁽¹⁰⁾

Reservoir

Forest dwelling fruit bats are thought to be the main reservoir for Ebola Virus.^(8,10) Non-human primates (e.g., gorillas, chimpanzees) and other wild animals (e.g., rodents, small antelopes) have been identified as intermediate hosts.⁽⁹⁾

Transmission

Person-to-person transmission can occur through: direct unprotected contact (via broken skin or mucous membranes) with blood or bodily fluids of a person that is sick with or has died from Ebola and/or objects that have been contaminated with body fluids (e.g. feces, blood, vomit) from a person sick with Ebola or from a person that has died from Ebola.^(7,10)

Transmission can also occur via direct contact with infected animals or their blood, organs or body fluids (i.e., ingestion or handling of dead or alive animals).^(7,10)

An extensive literature review to evaluate the evidence on modes of Ebola virus shedding and transmission showed viral persistence for nine months or more in a small percentage of EVD survivors.⁽¹²⁾ Ebola virus has been found in testicles/semen, vaginal fluid, placenta, amniotic fluid, human milk, cerebral spinal fluid, saliva, joints, conjunctiva, and chambers of the eye even after acute illness.^(7,8,12) The only documented cases of secondary transmission from recovered patients has been through sexual transmission.⁽¹²⁾ However, more research and surveillance data is needed to determine the risk of sexual transmission.⁽⁷⁾

For individuals infected while pregnant, Ebola virus persists in the placenta, amniotic fluid, and fetus.

For individuals infected while breastfeeding, Ebola virus may persist in breast milk.⁽⁷⁾

Ebola virus has the potential to be used as a bioterrorist agent.⁽¹³⁾

Incubation Period

Symptoms take between two to 21 days to appear, but usually present within four to nine days.^(6–8)

Period of Communicability

Infected individuals become infectious with onset of symptoms.⁽⁸⁾ Risk of transmission is highest when a person is acutely unwell and the viral load is at its peak. Transmission can occur as long as the virus is present in blood and body fluids.⁽⁷⁾ This includes the post-mortem period.⁽¹⁰⁾

Some body fluids (e.g., semen, urine, and breast milk) may contain Ebola virus for weeks to months following recovery from acute illness.

Host Susceptibility

Susceptibility is universal. The following people are at increased risk of exposure, where EVD transmission is active:^(8,14)

- caregivers (personal caregivers, family),
- healthcare workers caring for infected patients,
- laboratory workers processing clinical specimens or working in research laboratories,
- individuals that come in contact with contaminated medical equipment (e.g., needles and syringes) that were improperly sterilized,
- individuals that wash and prepare deceased bodies for burial,
- individuals that work closely with animals, particularly bats or non-human primates in Central or West Africa,
- individuals that work in bat-inhabited locations (e.g., mines).

Treatment

An infectious disease specialist should be consulted. To date, no specific therapeutics for EVD have been approved by Health Canada. Several investigational therapeutics (i.e., antivirals and monoclonal antibodies) are currently under development. The World Health Organization (WHO) published guidance on [Therapeutics for Ebola virus disease](#), recognizing that access could be challenging in resource-poor areas.⁽¹⁵⁾

If a confirmed case of EVD occurs in Canada, the PHAC will coordinate with the province or territory to provide guidance to the treating physician regarding access to investigational products, including monoclonal antibodies and vaccine, through the appropriate regulatory mechanism.⁽¹⁰⁾ [Ebola Clinical Care Guidelines](#) are available to support clinicians in Canada.

Incidence

EVD occurs mainly in areas surrounding rainforests in central Africa. There have been several outbreaks in a number African countries over the past few decades, some being notable for their high mortality rates. To date, no cases have been reported in Canada.⁽¹⁰⁾ More information on global current outbreaks can be found [here](#). In Alberta, EVD has been notifiable since 1985.

Public Health Management

Key Investigation

- Ensure person meets case definition for confirmed, probable or suspect case.
- Obtain history of illness including date of onset of signs and symptoms.
- Determine occupation (e.g., laboratory worker, humanitarian worker).
- Determine level of exposure within 21 days before symptom onset, including settings and activities while residing or traveling, in an area where EVD transmission is active. See [Table 1:Exposure Categories..](#)
- Inquire about ingestion or handling of infected animals (dead or alive).
- Determine possible transmission settings (e.g., household, school/childcare, healthcare, community) where other humans or animals (e.g., pet, farm, zoo animals) may have been exposed (see [Table 1:Exposure Categories](#)) to case.
- Identify contacts with high-risk exposure as per [Table 1: Exposure Categories.](#)

Table 1: Exposure Categories^(15,16)

Level of Risk	Type of Exposure
High-risk exposure	• Direct contact with the body surface/mucous membranes of a symptomatic EVD case or an ill individual in an affected area, their blood, body fluids or deceased body or contaminated surfaces or equipment, without adhering to recommended Infection, Prevention and Control (IPC) precautions or had a known IPC breach • Contact with contaminated medical instruments, objects, or environmental surfaces • Unprotected sexual contact with an acute or convalescent (12 months from onset of symptoms) EVD case • Pregnancy (mother to child) • Consuming breastmilk from a symptomatic EVD case or a recently recovered case
Low-risk exposure	• Direct contact with a symptomatic EVD case, their blood, body fluids or deceased body while adhering to recommended Infection Prevention and Control (IPC) precautions and with NO known IPC breaches • Living in the same household with a symptomatic EVD case but NO direct contact with their blood, body fluids or contaminated surfaces. • Casual interactions (e.g., sharing a seating area on public transportation or sitting in the same waiting room) with a symptomatic EVD case but NO direct contact with their blood or body fluids.

Management of a Returning Traveler

(Interim guidance during active outbreaks occurring outside of Canada)

- Travelers returning from an EVD affected area will be assessed by CBSA upon arrival. Information on EVD and interim guidance for returning travelers during an active outbreak of EVD outside of Canada is provided [here](#). Recommendations may include self-monitoring and notification to public health if indicated.

Management of a Case

- Symptomatic suspect, probable or confirmed cases must be isolated using, contact, droplet, and routine isolation precautions and managed in a hospital setting as per IPC precautions.⁽¹⁷⁾
 - Assessment and stabilization of confirmed cases will occur in Alberta as per AHS protocol and in consultation with the MOH and CMOH. Transfer to a dedicated treatment site (in Surrey, British Columbia. or Winnipeg, Manitoba) may be considered.
- Provide information about disease transmission and measures to minimize transmission, including practicing proper hand hygiene, avoiding physical contact with pets or other animals, and ensuring that animals or other people do not come in contact with their blood or body fluids (including urine, feces, emesis, saliva, sweat, and semen) or anything that may have come in contact with their blood or body fluid (e.g. linens, clothing, toilet, toiletries).

- Management of a convalescent confirmed case should be done on a case by-case basis in consultation with an infectious disease specialist and Public Health. Include counselling on the risk of transmission of EVD through body fluids such as semen and breast milk. Based on present evidence:
 - It is recommended that survivors of EVD born male should either abstain or use condoms correctly and consistently and have semen tested. If semen testing is not done, male cases should continue to use condoms or abstain from sexual activity for 12 months from onset of symptoms.⁽⁷⁾
 - It is recommended that individuals with active EVD and those who recently recovered from EVD should not breastfeed unless breastmilk tests negative for Ebola virus.⁽¹⁸⁾ Testing breastmilk for presence of EBV virus is recommended to help determine when breastfeeding can be safely resumed.⁽¹⁹⁾

Management of Deceased Cases

- All remains of deceased cases of EVD must be handled in accordance with Alberta's [Bodies of Deceased Persons Regulation](#). For additional information refer to the [Infection Prevention and Control Measures for Ebola Virus Disease in Healthcare Settings](#).

Management of Contacts

- Due to the severity of EVD and the lack of treatment options, a robust public health response to identify contacts of confirmed and probable cases must occur promptly.
- See [Table 2: Management of Asymptomatic Contacts of a Confirmed or Probable Case](#) and [Table 3: Management of Symptomatic Contacts of a Confirmed or Probable Case](#) for management of symptomatic and asymptomatic contacts.

Table 2: Management of Asymptomatic Contacts of a Confirmed or Probable Case^(15,16)

Exposure Category	Management
Low-risk Exposure (As defined by Table 1: Exposure Categories)	• Educate on signs and symptoms, • Monitor for EVD symptoms for 21 days after last possible exposure. If symptoms develop, isolate immediately (i.e., maintain a 2-meter distance and no physical contact with people or pets/animals). If medical attention is required, advise to call Health Link at 811 for further direction on where to go for care (indicating they are a contact of EVD), the appropriate mode of transportation to use, and Infection Prevention Control (IPC) precautions to be followed, • During the 21-day monitoring period, advise contacts to: - Avoid taking any antipyretic medication, if possible, as these may mask potential symptoms, - Postpone elective medical visits and other elective procedures (e.g., elective dental visits, elective blood tests). Any required medical assessments/appointments or procedures require advance notice of the MOH, VOC and clinic so that it can be performed safely, - Refrain from donating blood, sperm and any other body fluid or tissue, - Maintain good respiratory and hand hygiene practices.
High-risk Exposure (As defined by Table 1: Exposure Categories)	• Include advice as above for management of low- risk close contacts • Remain near an acute care facility (e.g., within one hour's drive, if possible, to facilitate rapid transfer to the facility) where medical care with appropriate IPC measures can be implemented • Either abstain from sexual contact or observe safe sex practices through correct and consistent condom use for 21 days • Quarantine (one incubation period) may be ordered by the direction of the Medical Officer of Health (MOH) in consultation with the CMOH.

Table 3: Management of Symptomatic Contacts of a Confirmed or Probable Case^(15,16)

Management
- Isolate immediately (i.e., maintain a 2-meter distance and no physical contact with people or pets/animals) if not already isolated from others. If medical attention is required, advise to call Health Link at 811 for further direction on where to go for care (indicating they are a contact of EBV), the appropriate mode of transportation to use, and Infection Prevention Control (IPC) precautions to be followed. - Wash hands, especially after vomiting or toileting, - Ensure that others do not come in contact with their blood or body fluids (including urine, feces, emesis, saliva, sweat, and semen) or anything that may have come in contact with their blood or body fluid (e.g., linens, clothing, toilet, toiletries). - Manage as Suspect Case. See Management of a Case

Post-Exposure Prophylaxis (PEP)

- Ebola Zaire vaccine is authorized in Canada and recommended for individuals 18 years or older for protection against EVD. This vaccine is not currently available or recommended for Canadians as part of routine immunization or for vaccination prior to travel⁽²⁾.
- EZV should be given within 72 hours of exposure but may be given up to 10 days post-exposure.⁽²⁾
- There is a limited supply of vaccine available as post-exposure prophylaxis to persons who have been exposed to EVD in Canada.
- The EZV vaccine is stockpiled in Canada's National Emergency Strategic Stockpile (NESS)⁽²⁰⁾ and is only available through the [Special Access Program \(SAP\)](#).
- In consultation with the CMOH, an MOH can seek authorization from Health Canada's SAP Program to access vaccine.
- The EZV vaccine is not indicated for use against other types of orthoebolviruses (i.e. Tai Forest, Sudan, Bundibugyo, Reston, and Bombali viruses) or related filoviruses⁽²⁾
- There are no licensed vaccines or therapeutics for the Sudan ebolavirus (SUDV).⁽²¹⁾

Management of Animal Contacts

- Consult with the Office of the Chief Provincial Veterinarian via the Public Health Veterinarian or designate when an animal (e.g., companion animal, livestock, exotic/zoo species) is identified as having had direct or indirect contact (e.g., prolonged close contact, contact with contaminated surfaces or items) with a probable or confirmed case while infectious so that a risk assessment can be performed and appropriate precautions implemented if required.

Management of Airplane Contacts

- Contact tracing of travelers on an airplane who may have been exposed to a case of EVD during a flight should be made on a case-by-case basis and consider the following:⁽²²⁾
 - the case's classification (i.e., confirmed, probable, suspect),
 - the type and severity of symptoms of the case during the flight, and
 - how long after the flight the case was identified.
- The risk of transmission from an infected symptomatic passenger is low in the early stages but may increase in the later stages of the disease, with increasing viral load and increased viral shedding.
- If a case (passenger) was symptomatic on the flight and the flight was within 21 days after last exposure to the case (see [Table 1:Exposure Categories](#)), contact tracing should focus on the following:
 - passengers seated within two meters of the index case,
 - crew and cleaning personnel serving the section of the aircraft where the index case was seated,
 - persons who had close contact with the index case (e.g., travel companions or persons providing care).

Preventive Measures

- Provide the following recommendations to persons travelling to an area where EVD is circulating:
 - Consult a health care provider or visit a travel health clinic before travel.
 - Refer to the Public Health Agency of Canada (PHAC) [Travel health Notices](#) and [Health advice for travellers](#).

Resources

- Alberta Health Services (AHS): [Ebola](#)
- Public Health Agency of Canada (PHAC): [Ebola](#)
- Centers for Disease Control and Prevention (CDC): [Ebola](#)
- World Health Organization (WHO): [Ebola](#)

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